## A Two perturbations of a frozen memory system

**STRUCTURAL ALIASING**

**Distinct inputs converge**

distinct cues → aliased cue

$\alpha \uparrow$ erases discriminative structure

**OBSERVATION NOISE**

**Structure survives perturbation**

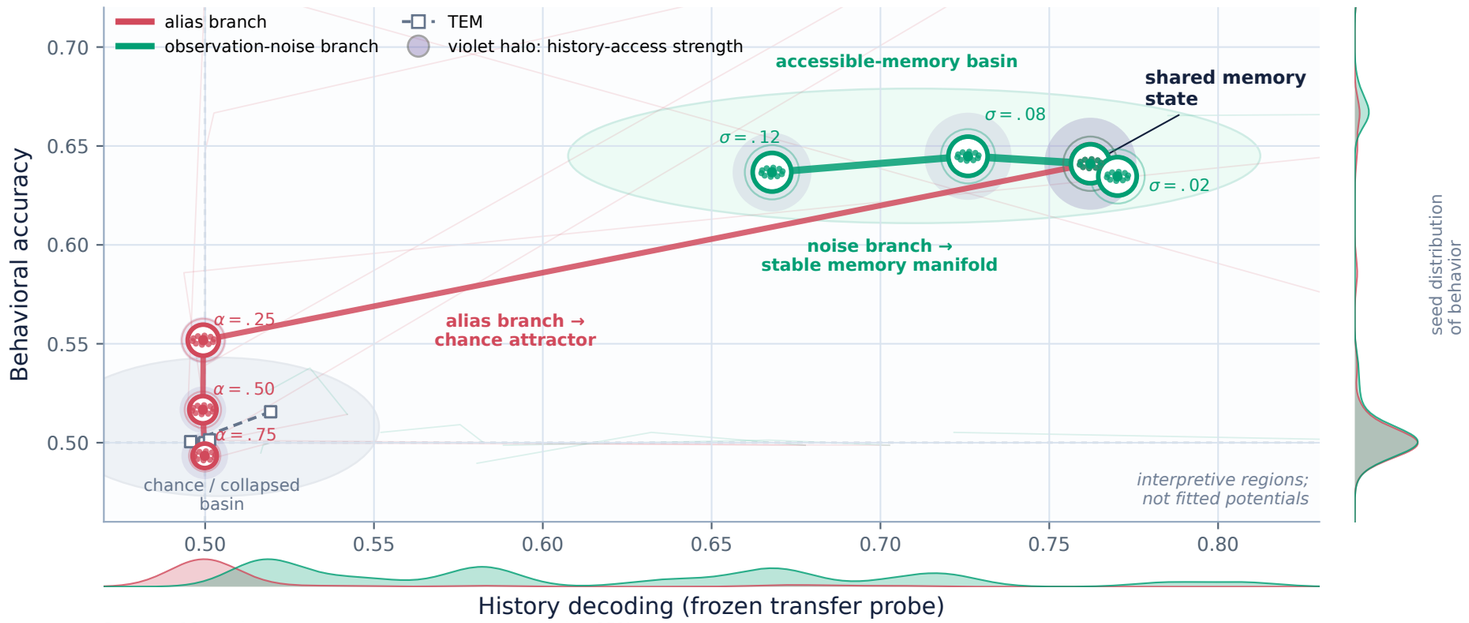
clean cue → noisy cue

$\sigma \uparrow$ adds stochastic variation

same frozen checkpoint → EC → CA3 → CA1 → choice

One factor varied at a time · cue cartoons are illustrative

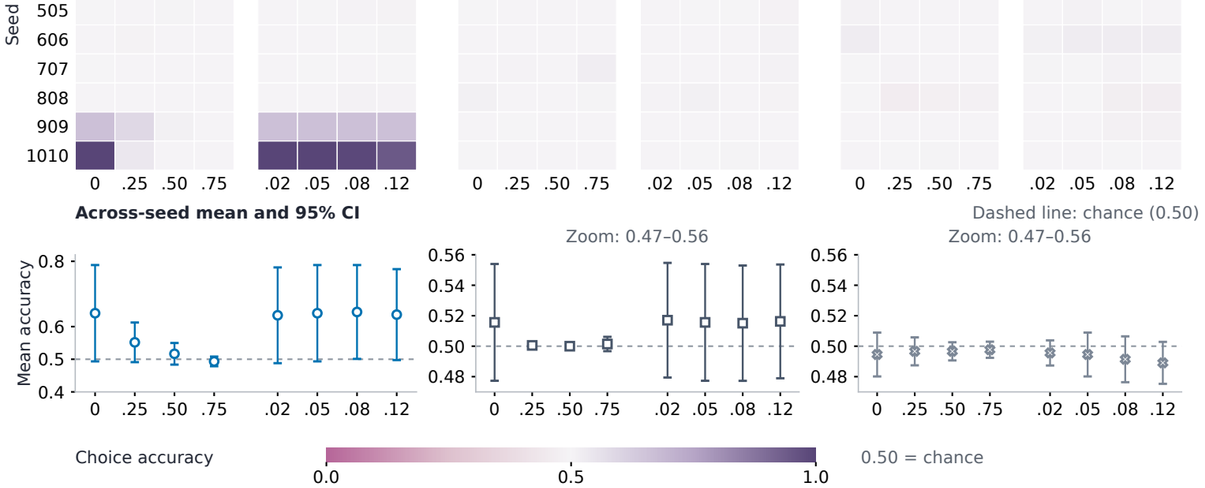
## B Hippocampal population bifurcation portrait



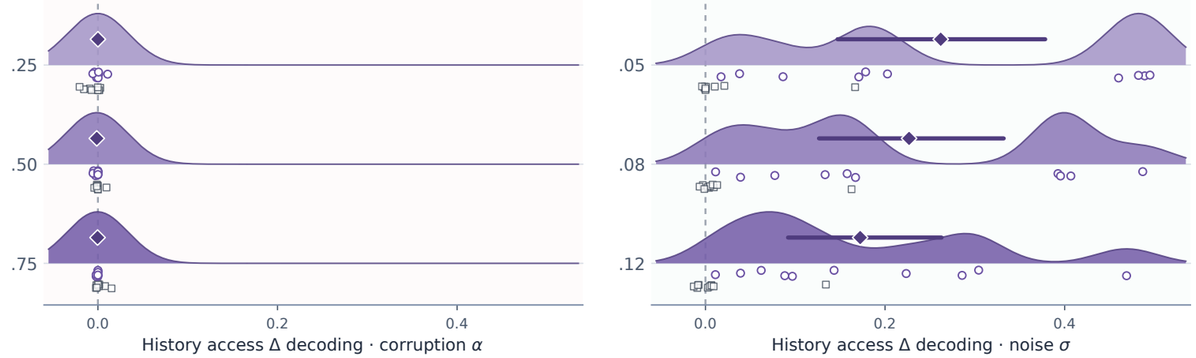

Packets: 10 seed marks; center: mean; envelope: relative 95% mean uncertainty.
Filaments: observed seed trajectories; trunk width: split-half stability. Main axes are zoomed.

## C Seed-resolved behavioral responses

Same seed order across models and perturbations

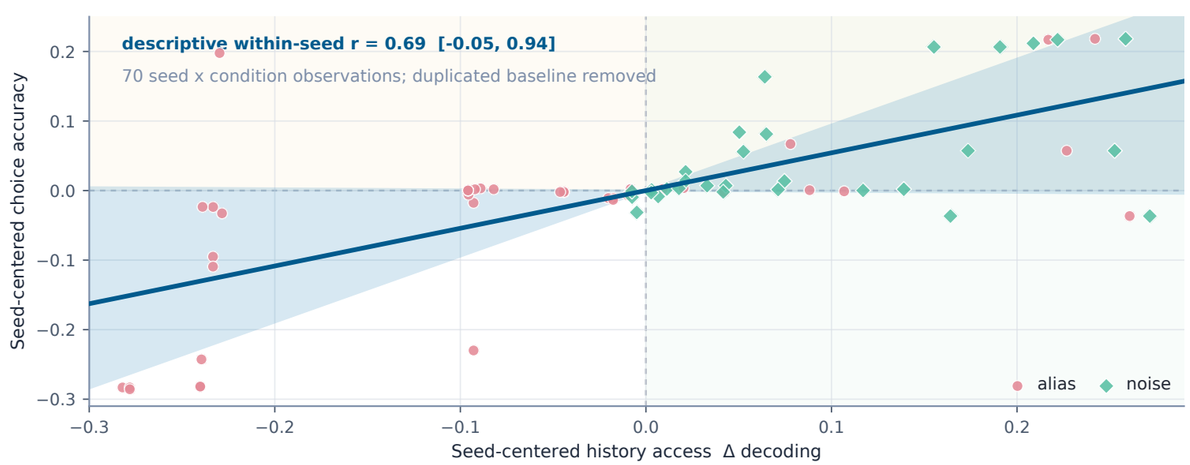



## D History access across perturbations

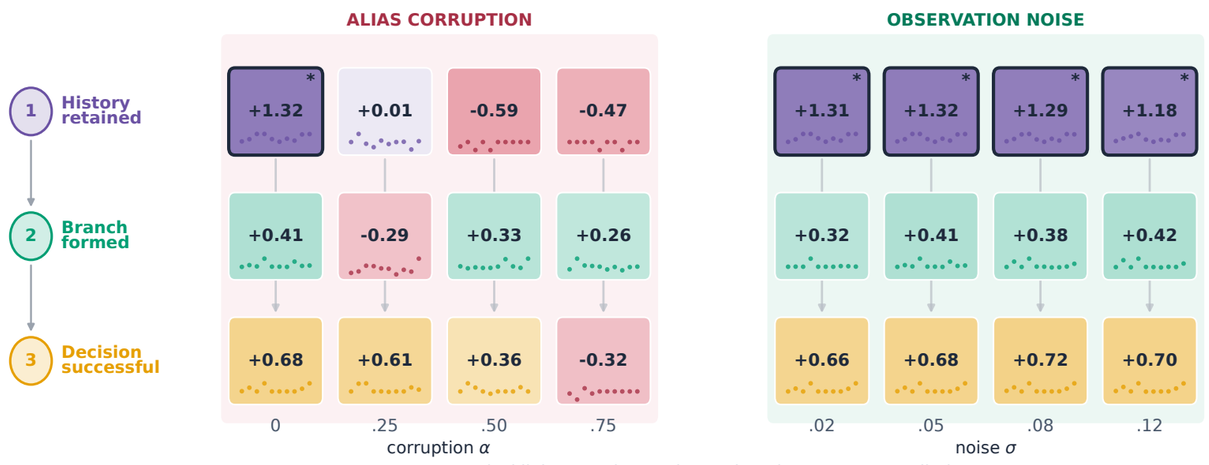



Open circles: HDGM seeds  Squares: TEM seeds
Diamonds and intervals: HDGM mean and 95% bootstrap CI

## E Within-seed access-behavior coupling




## F Readout effects across perturbations




vertical links organize readouts; they do not assert mediation
node = paired $d_z$ · 10 ticks = seeds · black border + * = Holm $p < .05$